pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

pandas: Series and DataFrames, Part 2

WEEK 2, DAY 2, PART 2

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

By the end of this part you can

Summarize a column

Totals, averages, counts, and shares.

```
holdings['sector'].value_counts()
```

Filter, count, and rank

By sector, rating, price, spread, or duration.

```
holdings[holdings['sector'] == 'Energy']
```

Add a column, run a what-if

On a copy, without touching the desk's record.

```
what_if = by_cusip.copy()
```

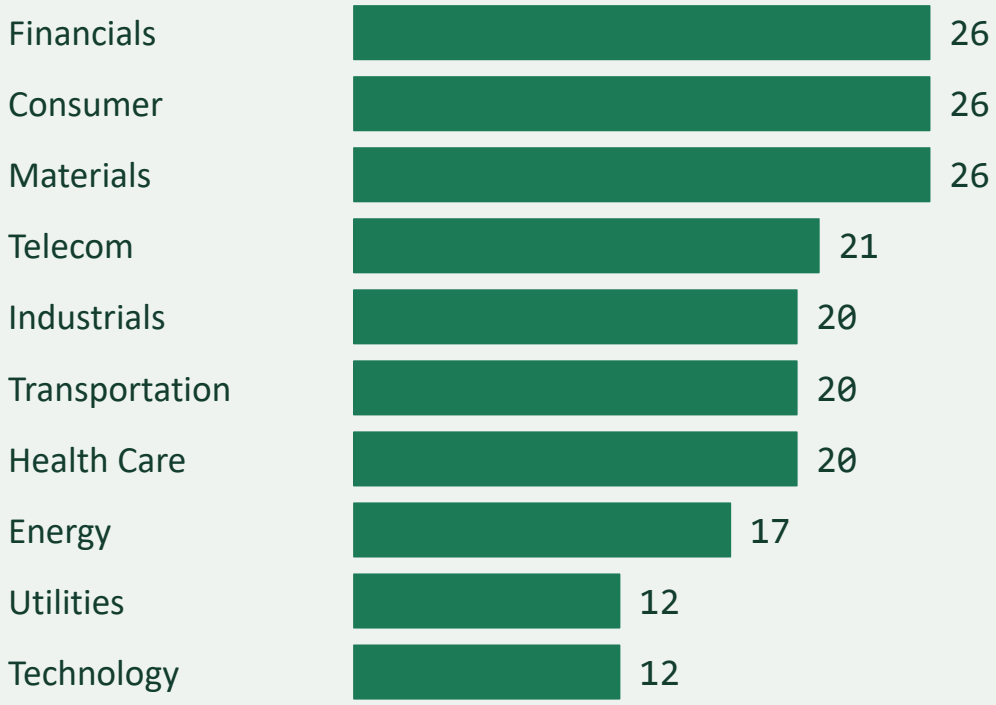
Show a table well

Formats, a caption, shading, and bars.

```
.style.format({'oas': '{:,.0f} bp'})
```

A column is a Series, so it can be summarized

`holdings['sector'].value_counts()`



A count of an invented portfolio. In Excel, a pivot table.

The same job as the Excel function, with the column named

pandas	Excel
`holdings['market_value'].sum()`	`=SUM(S2:S201)`
`holdings['oas'].mean()`	`=AVERAGE(M2:M201)`
`holdings['oas'].median()`	`=MEDIAN(M2:M201)`
`holdings['price'].min()`	`=MIN(I2:I201)`

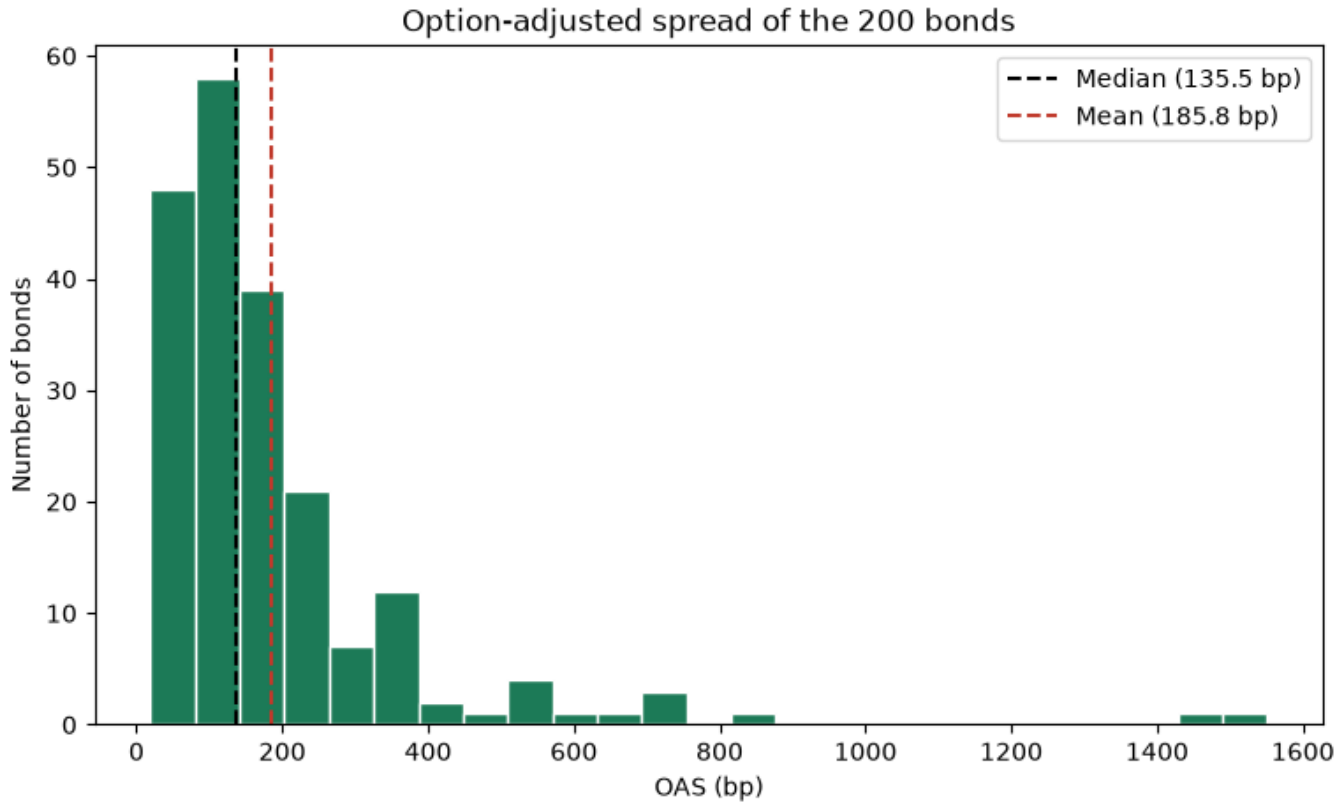
115 `nunique()`

tickers. Many issuers have more than one bond.

200 `nunique()`

CUSIPs for 200 rows. Every label is different.

Mean and median, one picture



Most bonds are tight

A tall block of bonds at spreads under 200 bp

A few are very wide

The tail runs out past 1,400 bp

The tail pulls the mean

Mean 185.8 bp, median 135.5 bp. Quote both.

Read the picture and check it against the numbers in the notebook.

A filter is a column of True and False

`holdings`

	ticker	sector
0	BEZE	Financials
1	BEZH	Industrials
...	...	...
198	ZEPN	Energy
199	ZEPN	Energy

the mask

is_energy
False
False
...
True
True

`holdings[is_energy]`

	ticker	sector
67	GALN	Energy
68	GALN	Energy
...	...	...
198	ZEPN	Energy
199	ZEPN	Energy

17

Energy bonds: the mask, summed

38,830,005

Their total market value

Question	pandas	Excel
Show the Energy bonds	`holdings[holdings['sector'] == 'Energy']`	AutoFilter on the sector column
Count the Energy bonds	`(holdings['sector'] == 'Energy').sum()`	=COUNTIF(E:E, "Energy")
Market value of the Energy bonds	`holdings.loc[is_energy, 'market_value'].sum()`	=SUMIF(E:E, "Energy", S:S)

& narrows. | widens.

& keeps the rows where both conditions are True

```
long_and_low = (holdings['duration'] > 10) & (holdings['price'] < 90)
print('Count:', long_and_low.sum())
Count: 6
```

6 bonds pass both tests

| keeps the rows where at least one condition is True

```
low_or_wide = (holdings['price'] < 80) | (holdings['oas'] > 1000)
print('Count:', low_or_wide.sum())
Count: 4
```

4 bonds pass at least one test

`( ... ) & ( ... )`

Each condition in its own round brackets.

`and, or`

Same ValueError as in NumPy. Write & and |.

`isin(['A+', 'A', 'A-'])`

True where the value is in the list. 56 single-A bonds.

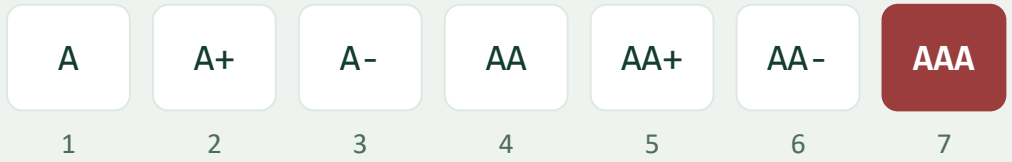
Sorting hands back a sorted copy

```
widest = holdings.sort_values('oas', ascending=False)
```

	ticker	sector	rating	price	oas
65	FESR	Technology	CCC+	87.116	1551
17	CLDB	Consumer	B	62.045	1446
132	PRAN	Financials	CCC+	89.407	871
110	NEVN	Utilities	B-	96.616	720
111	NEVN	Utilities	B-	94.053	719

The labels moved with their rows: the first row is still labelled 65. holdings itself is not sorted. The issuers are invented.

Sort by rating, and text sorts letter by letter



Alphabetical order, not credit order. AAA comes out seventh.

```
holdings.nlargest(5, 'market_value')
```

A shortcut for sort, then head.

In Excel: Data, Sort

Excel sorts the sheet in place. pandas leaves the original order alone.

A new column is a formula column

```
holdings['coupon_income'] = holdings['par_held'] * holdings['coupon'] / 100
```

	ticker	coupon	par_held	coupon_income
0	BEZE	7.000	5000000	350000.0
1	BEZH	8.125	250000	20312.5
2	BEZL	5.125	1500000	76875.0

In Excel

`=R2*G2/100`

A formula in a new column, filled down. A pandas column holds values, not live formulas: change an input, and rerun the line.

1 Work on a copy

```
what_if = by_cusip.copy()
```

2 Set one value with loc

```
what_if.loc['99039NAC0',
            'price'] = 97.0
```

3 Recalculate, then compare

```
print('Change in market value:',
      round(change, 2))
Change in market value: -25950.0
```

Change many rows at once, on a copy

loc[mask, column] = value sets many rows

```
marks = holdings.copy()
marks['price_check'] = 'OK'
marks.loc[marks['price'] < 80,
          'price_check'] = 'Confirm'
Rows marked Confirm: 3 of 200
```

CLDB at 62.045, NEVX at 73.873, SYLX at 76.903. The other 197 rows still read 'OK'.

Filter, then .copy(), before you change the table

```
energy_what_if = holdings[is_energy].copy()
energy_what_if['price'] = energy_what_if['price'] - 2
Average Energy price in the what-if:  95.407
Average Energy price in the holdings: 97.407
```

The holdings did not move. With .copy() the code gives the same result in every version of pandas.

Step	pandas	Excel
Set where a test holds	`marks.loc[marks['price'] < 80, 'price_check'] = 'Confirm'`	=IF(I2<80, "Confirm", "OK") filled down
Copy the filtered rows	`holdings[is_energy].copy()`	AutoFilter, copy the rows shown, paste

.style formats a table for display

Format, title, shade, and bar. The numbers underneath do not change.

```
(top10.style
  .format({'price': '{:.2f}',
           'oas': '{:,.0f} bp'})
  .background_gradient(
      subset=['oas'], cmap='Greens')
  .highlight_max(subset=['price'])
  .set_caption('Ten widest spreads'))
```

A cell still holds 106.727 when Format Cells shows 106.73. Style changes the display only.

Each method has an Excel twin

pandas	Excel
`.format()`	Format Cells
`.background_gradient()`	Conditional Formatting, Color Scales
`.bar()`	Conditional Formatting, Data Bars
`.highlight_max()`	Top/Bottom Rules, Top 1
`.set_caption()`	a title above the table

The AI check

Every notebook ends with code written in the style of an AI assistant. It has one bug. Find it.

1

Read

Read the code before you run it.

2

Predict

Decide what it should produce.

3

Run

Run it and look at the output.

4

Check

Compare with a number you worked out yourself.

Code that runs is not the same as code that is right.

What you can do now

- Total, average, and count a column, and read the share of each group
- Filter by sector, rating, price, spread, or duration, and rank what passes
- Add a column, change rows where a test holds, and run a what-if on a copy
- Drop and renumber, and show a table with formats, shading, and bars

Tomorrow pandas: joining tables, reading and saving files, summaries by group, and dates